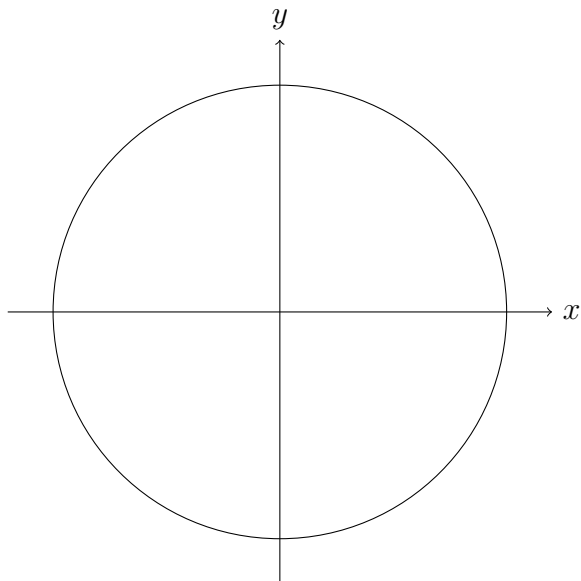


1 Revision

1.1 Symmetry properties



- $\cos(-x) = \cos(x)$
- $\sin(-x) = -\sin(x)$
- $\tan(-x) =$
- $\sin(x + \pi) =$
- $\cos(x + \pi) =$
- $\sin(x + \frac{\pi}{2}) =$
- $\sin(x - \frac{\pi}{2}) =$
- $\cos(x + \frac{\pi}{2}) =$
- $\cos(x - \frac{\pi}{2}) =$

1.1.1 Pythagorean identity

- $\sin^2(x) + \cos^2(x) =$
- $1 + \tan^2(x) =$
- $1 + \cot^2(x) =$

1.2 Solving equations

Strategy:

1. Find two distinct solutions in $[0, 2\pi)$ (or $[-\pi, \pi)$)
2. Use periodicity to find the general solution

Example questions:

1. Solve $\sin(2x) = \frac{\sqrt{3}}{2}$
2. Solve $(\sin(2x) - 1)(\sqrt{3}\sec(\frac{x}{5}) - 2) = 0$

1.3 Compound angles

- $\sin(a + b) =$
 - $\cos(a + b) =$
 - $\tan(a + b) =$
 - $\sin(a + b) + \sin(a - b) =$
 - $\cos(a + b) - \cos(a - b) =$
 - $\tan(2x) =$
1. let $\sec(x) = -3$, $x \in [\pi, \frac{3\pi}{2}]$ find the value of $\cot(2x)$
 2. Find the value of $\tan(\frac{\pi}{8})$
 3. Find without calculus the max and min values of $\sqrt{4\sin^2(t) + 36\cos^2(2t)}$
 4. Given that $\operatorname{cosec}(x - y) = \frac{5}{4}$, $\tan(x)\cot(y) = 10$ What is $\sin(x + y)$?